18 Make m the subject of

$$f = \frac{4 - 3m}{5 + m}$$

$f(5 + m) = 4 - 3m$

(+3m) (+3m)

$5f + mf + 3m = 4$

(−5f) (−5f)

$mf + 3m = 4 - 5f$

1 mark
Multiply by 5 + m

1 mark
Gather m terms

×	f	+3
m	mf	+3m

$m(f + 3) = 4 - 5f$

(÷(f + 3)) (÷(f + 3))

1 mark
Factorise m

$$m = \frac{4 - 5f}{f + 3}$$

1 mark
m = (4 − 5f)/(f + 3)

$f \neq -3$ (denominator must be nonzero).

(Total for Question 18 is 4 marks)